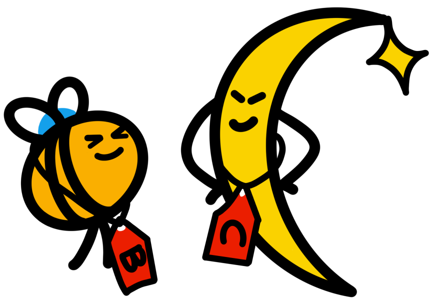
Documentation for preliminary 2D pseudocontinous SFs



AN-25-222

- Glance (potentially not the most up-to-date, but it will be coherent)
- GitLab (most up-to-date version, but cannot promise it is 100% coherent)

Reports in BTV meetings

- 2026.05.26
- 2025.10.27

Discriminator definitions

- **score(B vs. C)**: `btagUParTAK4BvC = 1 - btagUParTAK4CvB`
- `btagUParTAK4S = btagUParTAK4SvUDG * btagUParTAK4UDG / (1.0 - btagUParTAK4SvUDG)`
- `btagUParTAK4C = btagUParTAK4CvL * (btagUParTAK4S + btagUParTAK4UDG) / (1.0 - btagUParTAK4CvL)`
- `btagUParTAK4B = (1.0 - btagUParTAK4CvB) * btagUParTAK4C / btagUParTAK4CvB`
- **score(HF vs. LF)**: `btagUParTAK4HFvLF = (btagUParTAK4B + btagUParTAK4C) / (btagUParTAK4B + btagUParTAK4C + btagUParTAK4S + btagUParTAK4UDG)`

Preliminary versions

2026.06.29

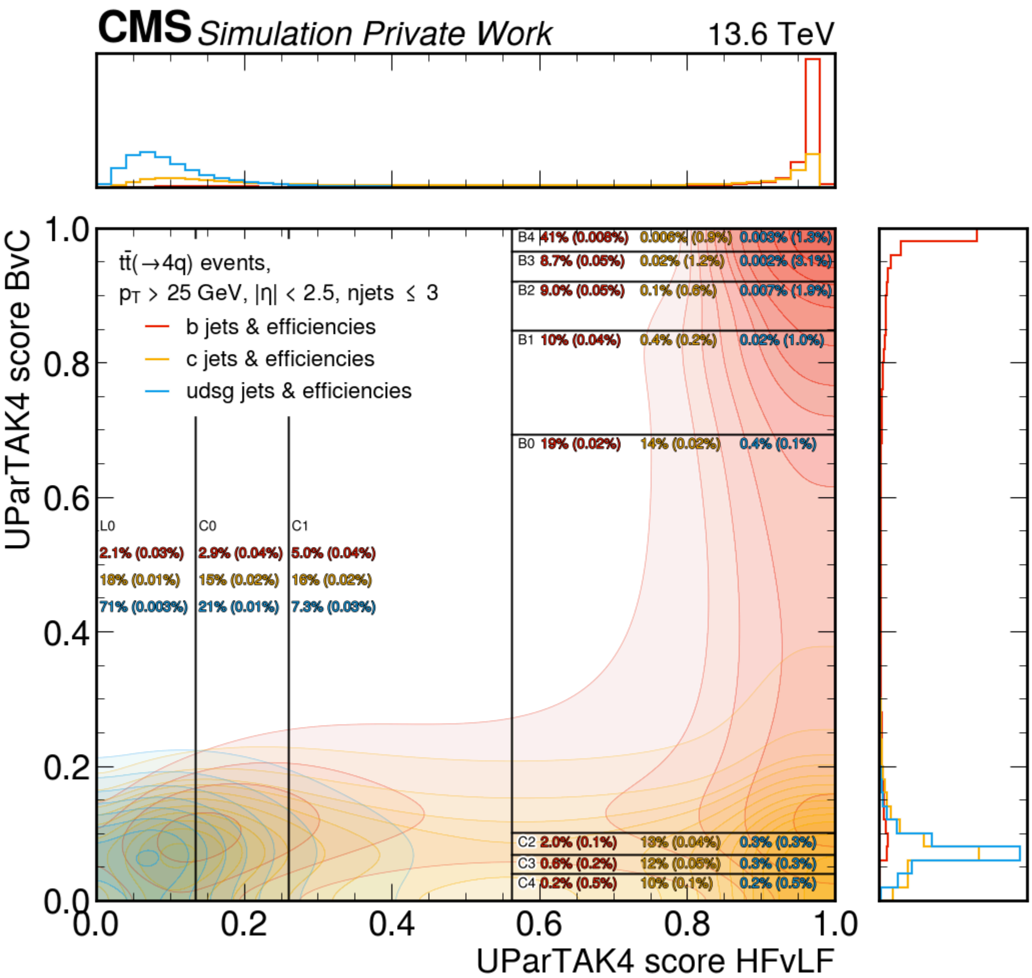
▶ Correctionlib printout (click to expand)

▶ Correctionlib printout (click to expand)

Tagger

UParT v2, available in NanoAODv15

Working point definitions



- HFvLF: `[0.000, 0.250, 0.452, 0.808, 1.000]`
- BvC: `[0.000, 0.006, 0.017, 0.055, 0.761, 0.944, 0.985, 0.995, 1.000]`
- Plots

▶ Samples (click to expand)

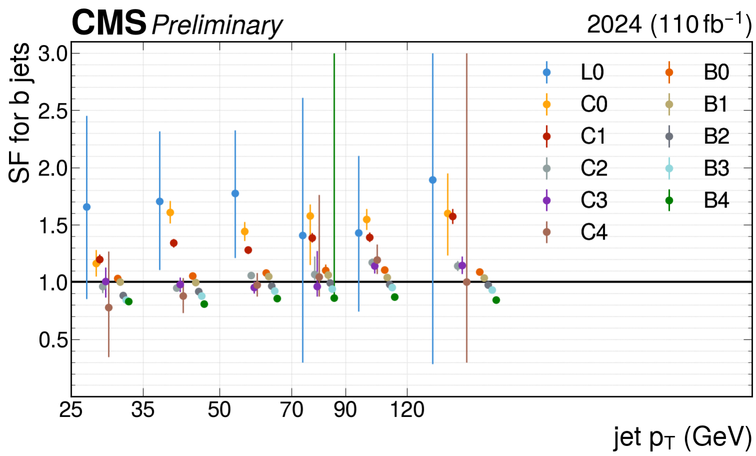
▶ Systematic uncertainties considered in the fit (click to expand)

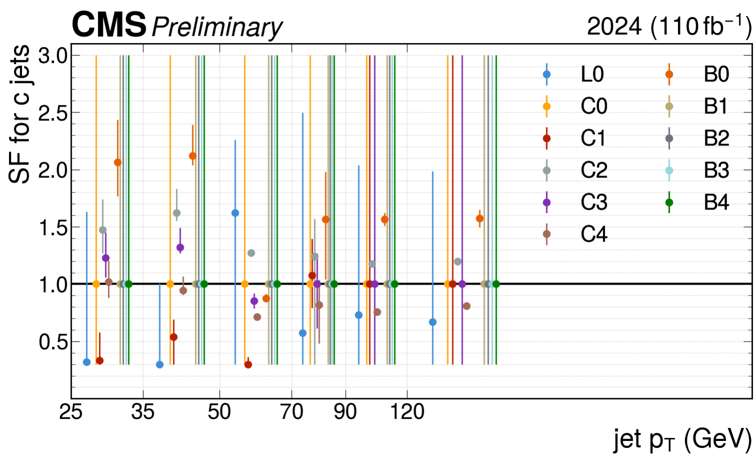
Notes

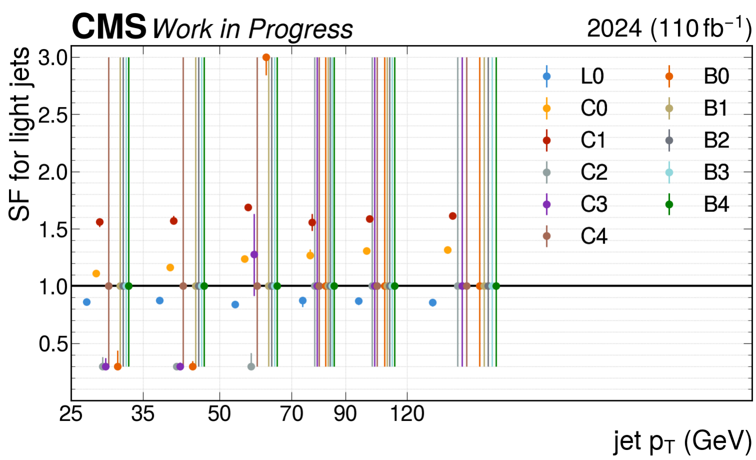
- Same Summer24 MC is used to derive 2024 and 2025 SFs, rather than split the MC into two halves
- Switch to NLO W+jets samples binned by both lepton flavour and number of jets
- 2025 SFs now mainly use 2025 corrections (except missing electron trigger SFs)
- Electron ID for 2025 is `wp80iso` , since `PromptMVA-Tight` is not yet available
- Add hdamp, bfragmentation, and cfragmentation uncertainties

Code

2026.04.28





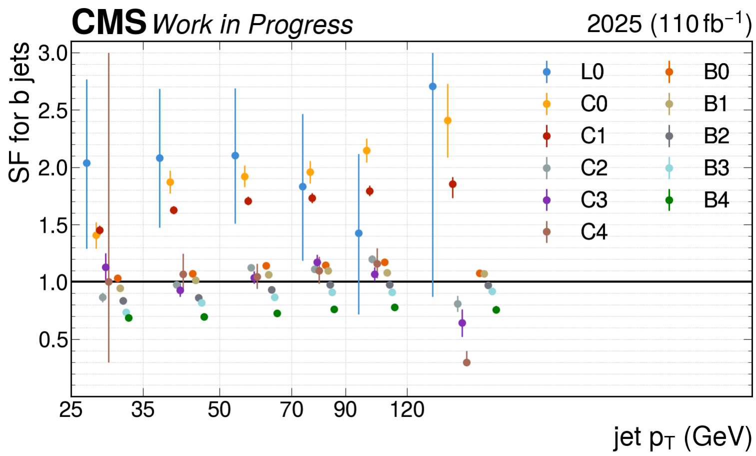


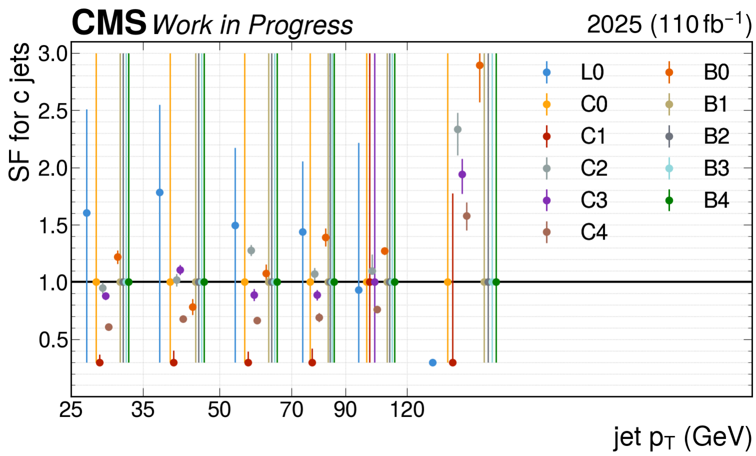
`/eos/cms/store/group/phys_top/Run3Vcb/flavTagSFs/20260428/flavTaggingSF_2024.json.gz`

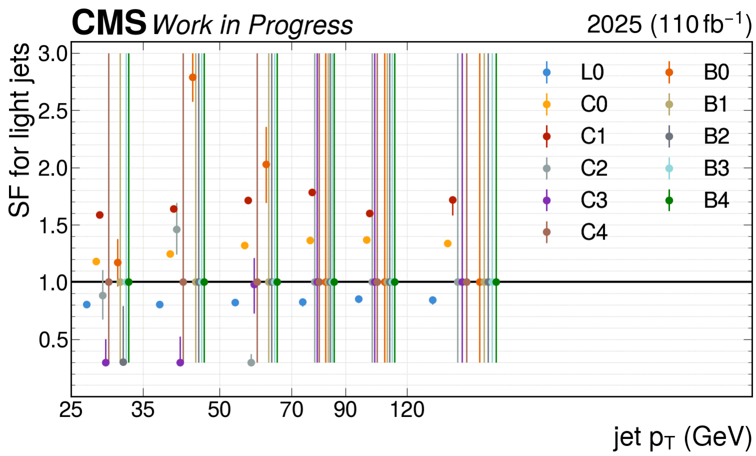
SF summary plots & impact plots & pre/postfit plots here

Template plots here

▶ Correctionlib printout (click to expand)







`/eos/cms/store/group/phys_top/Run3Vcb/flavTagSFs/20260428/flavTaggingSF_2025.json.gz` (very preliminary)

SF summary plots & impact plots & pre/postfit plots here

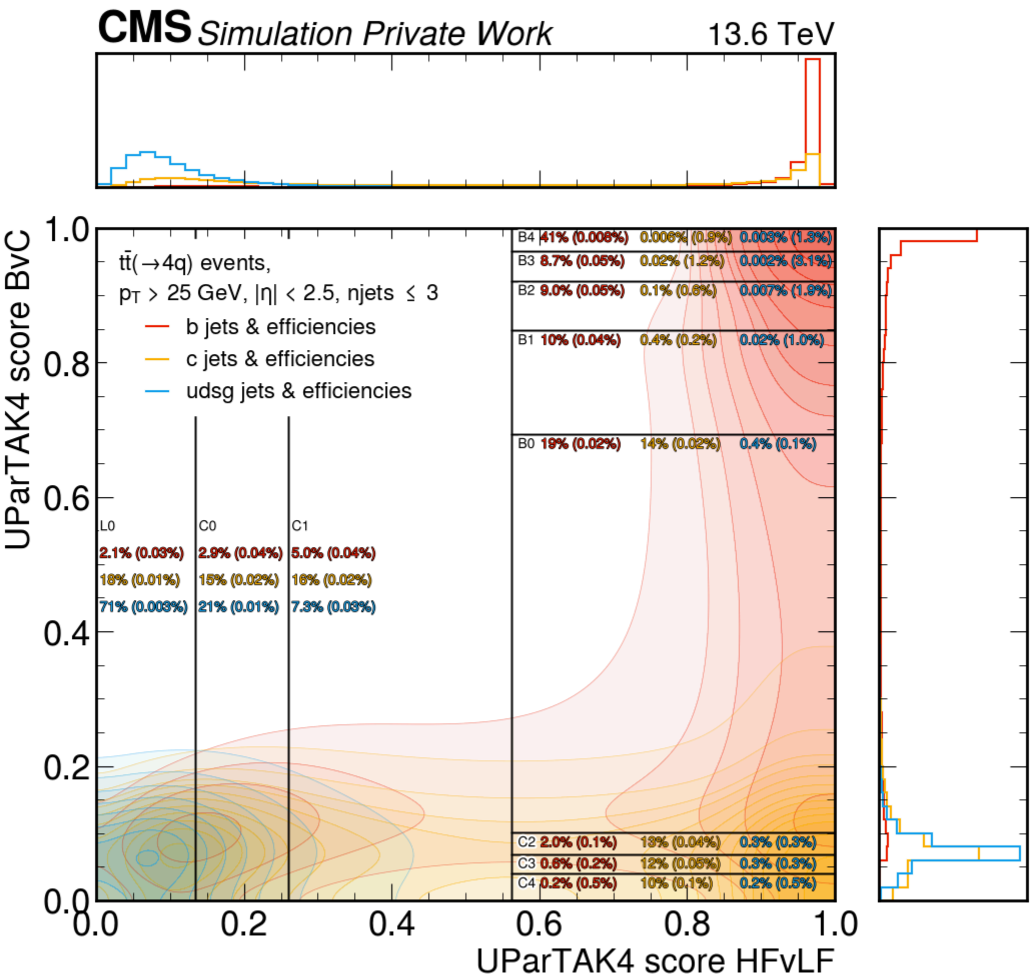
Template plots here

▶ Correctionlib printout (click to expand)

Tagger

UParT v2, available in NanoAODv15

Working point definitions



- HFvLF: `[0.000, 0.250, 0.452, 0.808, 1.000]`
- BvC: `[0.000, 0.006, 0.017, 0.055, 0.761, 0.944, 0.985, 0.995, 1.000]`
- Plots

▶ Samples (click to expand)

▶ Systematic uncertainties considered in the fit (click to expand)

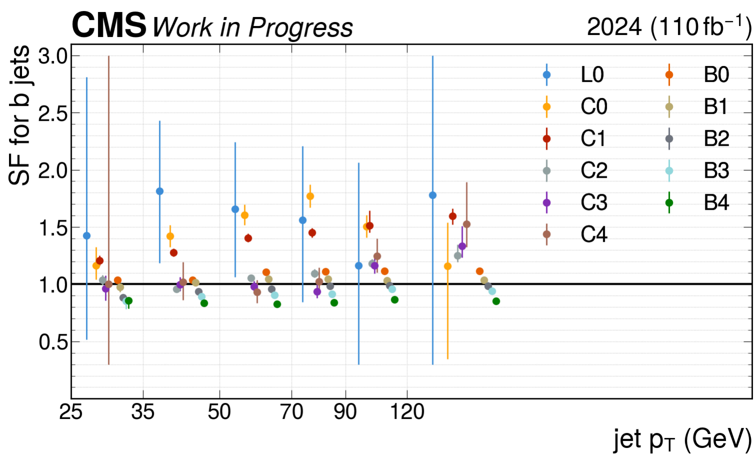
Notes

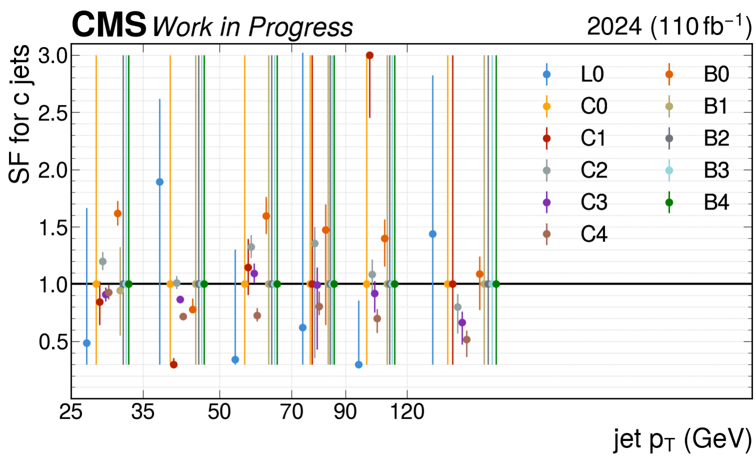
- Switch to NLO W+jets samples (only binned by lepton flavour, not binned by number of jets)
- Switch to NLO low mass DY samples
- Added `WZtoL3Nu` sample
- 2025 SFs mainly use 2024 corrections
- Same Summer24 MC is used to derive 2024 and 2025 SFs, rather than split the MC into two halves
- Shape (W+jets) and lnN (tt dileptonic and Z+jets) trigger SFs are decorrelated in the fit
- Switched all phase spaces to use PromptMVA ID for leptons (previously only tt dileptonic used PromptMVA)

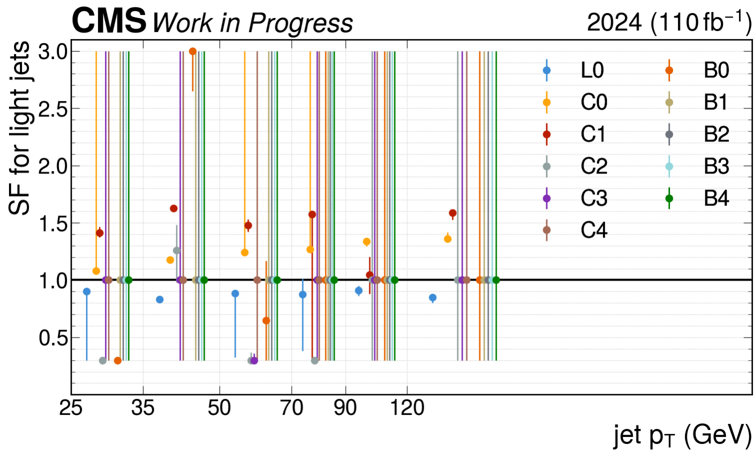
Code

- BTVNanoCommissioning commit
- SFbc-2D commit

2026.03.27







/eos/cms/store/group/phys_top/Run3Vcb/flavTagSFs/20260327/flavTaggingSF_2024.json.gz

SF summary plots & impact plots & pre/postfit plots here

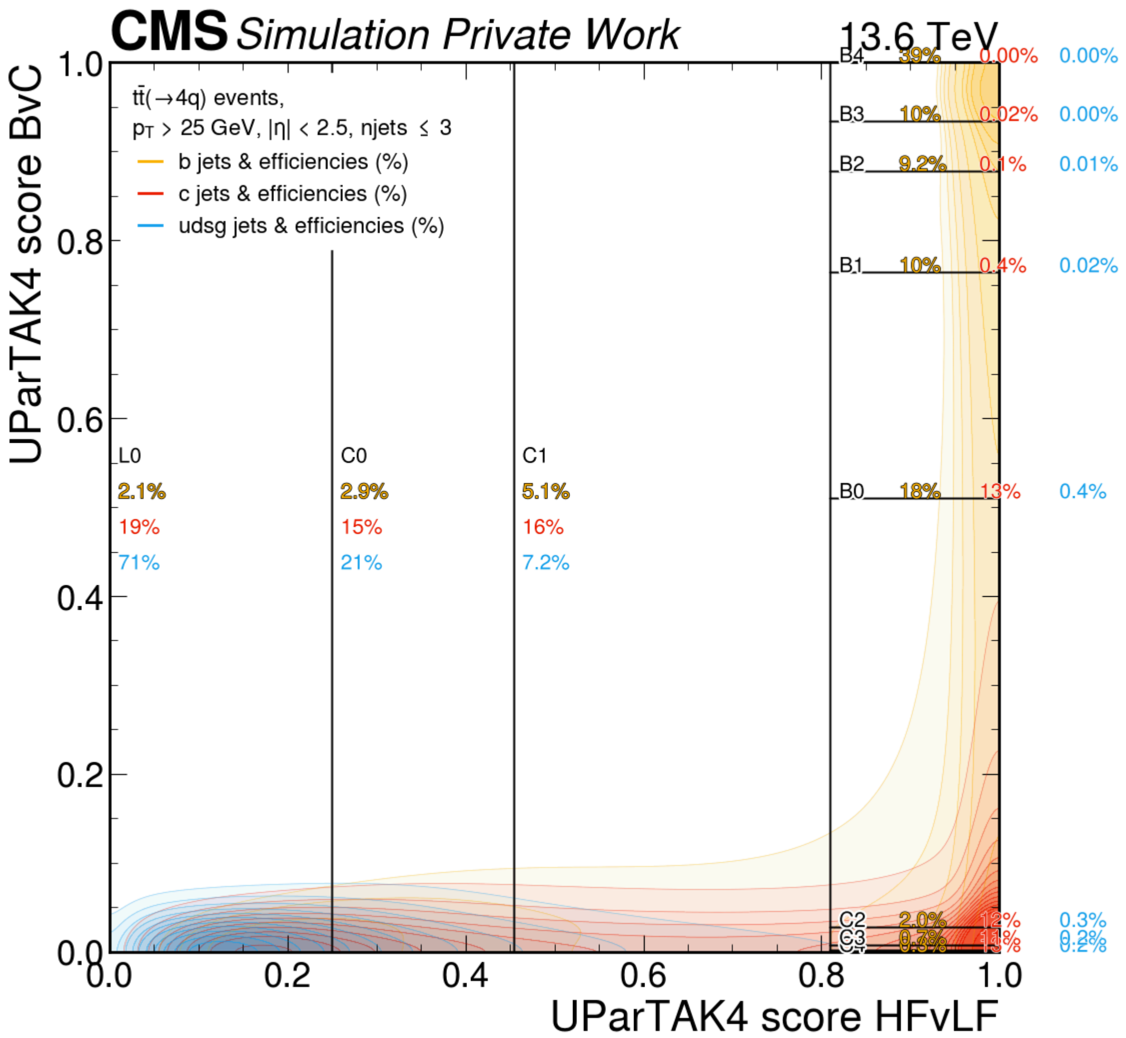
Template plots here

▶ Correctionlib printout (click to expand)

Tagger

UParT v2, available in NanoAODv15

Working point definitions



- HFvLF: `[0.0, 0.250, 0.454, 0.810, 1.0]`
- BvC: `[0.0, 0.006, 0.016, 0.056, 0.760, 0.944, 0.984, 0.994, 1.0]`
- Plots

▶ Samples (click to expand)

▶ Systematic uncertainties considered in the fit (click to expand)

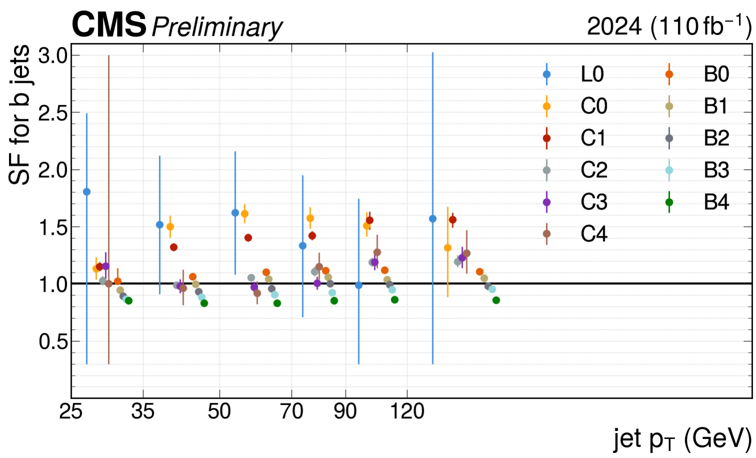
Notes

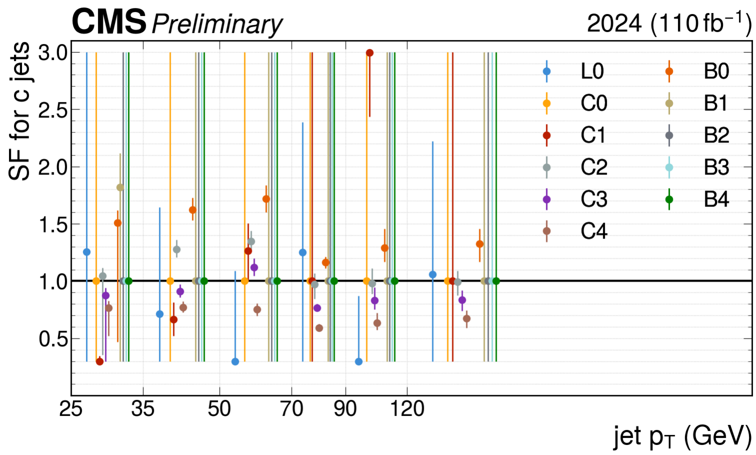
- Added `WZToLNu2B` sample
- Muon ID uncertainty is missing, instead the muon isolation uncertainty is present twice
- Switched to reduced split JES
- Fixed top pT weight to only be applied once
- Added trigger, MET unclustered energy, PDF, alphaS uncertainties
- PromptMVA nuisances decorrelated from wp08iso/pfRelIso04All nuisances

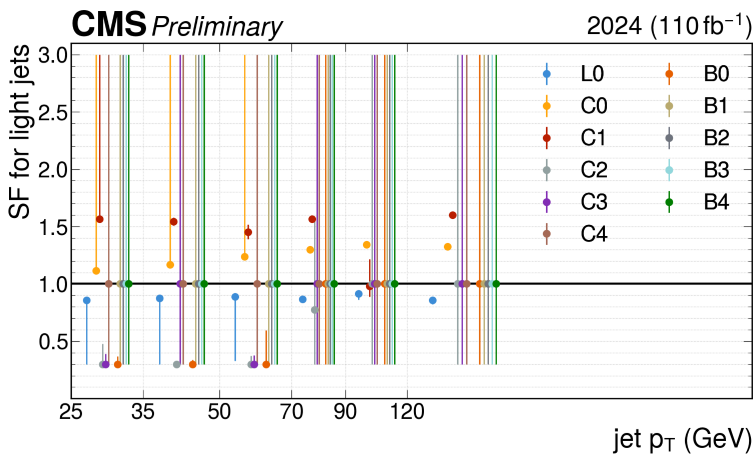
Code

- BTVNanoCommissioning commit
- SFbc-2D commit

2026.02.27







`/eos/cms/store/group/phys_top/Run3Vcb/flavTagSFs/20260227/flavTaggingSF_2024.json.gz`

SF summary plots & pre/postfit plots here

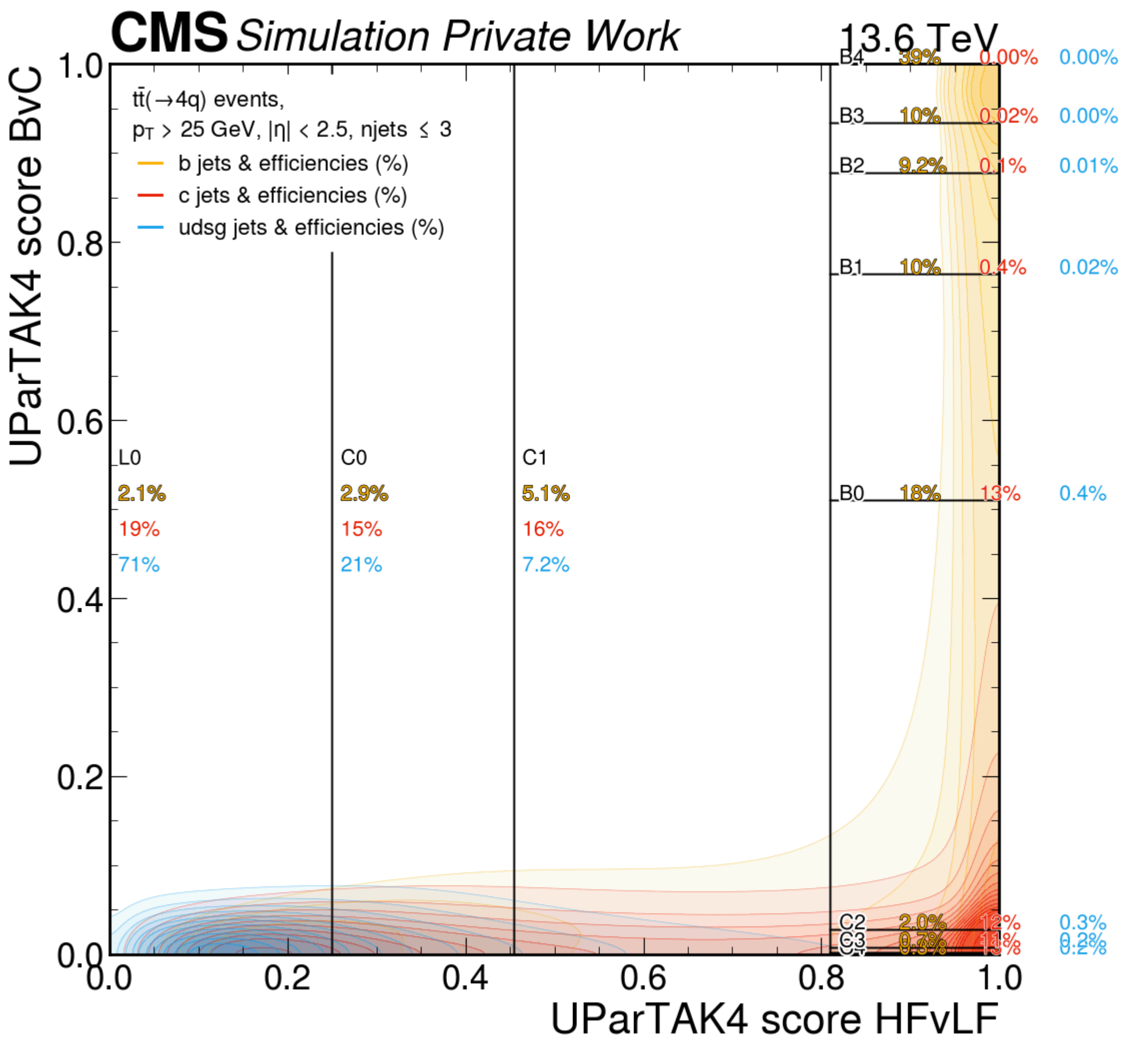
Datacards & impacts plots here

▶ Correctionlib printout (click to expand)

Tagger

UParT v2, available in NanoAODv15

Working points



- HFvLF: `[0.0, 0.250, 0.454, 0.810, 1.0]`
- BvC: `[0.0, 0.006, 0.016, 0.056, 0.760, 0.944, 0.984, 0.994, 1.0]`
- Plots

▶ Samples (click to expand)

▶ Systematic uncertainties considered in the fit (click to expand)

Notes

- First version with full set of uncertainties
- Top pT weight is applied twice

Code

- BTVNanoCommissioning commit
- SFbc-2D commit